

Geometry & Topology II: Course Notes

Mathilda Campillo

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Remarks: These are the notes for Geometry and Topology II, Spring 2026. My personal comments will be in [teal](#).

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1 Vector bundles

1.1 Dual vector bundle

1.2 Cotangent bundle

Definition: The cotangent bundle is defined as

$$T^*M \equiv (TM)^*.$$

We call elements of this dual space, *covectors*. What we are doing here is just taking the dual vector space to T_pM at each point $p \in M$. As in the case of the tangent bundle, smooth local coordinates for M yield smooth local coordinates for its cotangent bundle

Recall: There is a one-to-one correspondence between between tangent vectors at p and derivations at p .

$$\{\text{vectors} \in T_pM\} \longleftrightarrow \{v : C^\infty(M) \rightarrow \mathbb{R} : v(fg)|_p = f|_p v(g) + g|_p v(f)\}$$

Notice that given a point $p \in M$, a covector $\omega \in T_p^*M$, and a tangent vector $v \in T_pM$, then one can define a pairing $\langle \omega, v \rangle \in \mathbb{R}$. This follows basically from the definition of dual vector space. Now, given a function f locally defined near $p \in M$ and a tangent vector $v \in T_pM$, one can similarly define a pairing given by $\langle df_p, v \rangle = v(f)|_p \in T_p^*M$, where v is considered as a derivation defined near $p \in M$. We now give the precise definition of the differential.

Definition: Let $f : U \rightarrow \mathbb{R}$ be a smooth function defined on an open neighborhood of $p \in M$. Then the differential (of f at $p \in M$) is defined as

$$\begin{aligned} df_p : T_pM &\rightarrow T_p^*M \\ v &\mapsto df_p(v) \equiv v(f)|_p \end{aligned}$$

The differential assigns a scalar to each tangent vector via the pairing in the previous comment.

1.2.1 How does df looks more concretely?

We can compute the differential in local coordinates as follows:

Given a local chart (U, x) near p , then each coordinate function x^i is smooth and has differential $dx^i|_p$. Moreover, the differentials $\{dx^1|_p, \dots, dx^n|_p\} \subset T_p^*M$ provide a dual basis to $\{\frac{\partial}{\partial x^1}|_p, \dots, \frac{\partial}{\partial x^n}|_p\} \subset T_pM$. In these local coordinates, we can write the differential as

$$df_p = \sum_i^n \frac{\partial f}{\partial x^i} \Big|_p dx^i \Big|_p$$

Notice that we can think of df as an analogue of the usual gradient, in a way that makes coordinate-independent sense on a manifold.

Lemma: $dx^i|_p, \dots, dx_n|_p$ is a basis for T_p^*M .

Some useful remarks:

1. Any covector $\omega_p \in T_p^*M$ can be written uniquely as

$$\omega_p = \omega_i dx^i \Big|_p, \quad \text{where } \omega_i = \omega \left(\frac{\partial}{\partial x^i} \Big|_p \right).$$

This secretly says that all covectors come from differentials.

2. For each smooth function $f : M \rightarrow \mathbb{R}$, the map

$$\begin{aligned} df : M &\rightarrow T^*M \\ p &\mapsto df_p \in T_p^*M \end{aligned}$$

is a smooth section. Why? because at each point we are choosing the linear functional $df_p : T_pM \rightarrow \mathbb{R}$. Now, we will later see that these sections will define something called covector fields or differential 1-forms.

Properties: Let $f, g \in C^\infty(M)$. Then

1. $d(af + bg) = a df + b dg$
2. $d(fg) = g df + f dg$
3. $F : N \rightarrow M$ smooth map between smooth manifolds,

$$d(f \circ F)(V) = df(DF(v)) \forall v \in T_q N, q \in N.$$

Slightly off topic sidenote: We know that geometrically, we can think of a vector field on M as a rule that assigns an arrow to each point of M , but what kind of geometric picture can we form of a covector field?

The take away idea is that a nonzero linear functional $\omega_p \in T_p^*M$ is completely determined by two things:

1. Its kernel, which is a codimension one linear subspace of T_pM
2. The set of vectors in T_pM such that $\omega_p(v) = 1$, which determines an affine hyperplane parallel to its kernel.

Then, we can visualize a covector field as a defining pair of hyperplanes in each tangent space, one through the origin and another parallel to it, varying continuously from point to point.

Notice that up to this point, we have defined two different kinds of derivatives of f at a point $p \in M$:

1. The pushforward $f_* : T_pM \rightarrow T_{f(p)}\mathbb{R}$.
2. The differential $df_p : T_pM \rightarrow \mathbb{R}$.

But these are the same locally, once we take into account the canonical identification between $\mathbb{R}$ and its tangent space at any point.

Lemma: $f \in C^\infty(M)$, $f : M \rightarrow \mathbb{R}$ smooth, $\gamma : [0, 1] \rightarrow M$ smooth path. Then

$$F(\gamma(1)) - F(\gamma(0)) = \int_0^1 df(\gamma'(t)) dt.$$

Definition: A 1-form is a smooth section ω of T^*M . If $F : N \rightarrow M$ is a smooth function, then the pullback of ω is $F^*\omega$ on N , satisfying

$$F^*\omega(v) = \omega(DF(v)), \quad \forall v \in T_p M.$$

Definition: $\gamma : [a, b] \rightarrow M$ be a smooth map. The line integral of ω along γ is

$$\int_a^b \omega(\gamma'(t)) dt.$$

In general: $\omega|_p = \sum \omega|_p \frac{\partial}{\partial x_i} dx^i$

Examples: (1) $M = \mathbb{R}^2$ and consider $\omega = xdy$ which is not exact (not even close). Take the path $\gamma : [0, \pi] \rightarrow \mathbb{R}^2$ given by $\gamma(t) = (\cos(t), \sin(t))$. Then

$$\gamma'(t) = \sin(t) \frac{\partial}{\partial x} + \cos(t) \frac{\partial}{\partial y}$$

$$\omega(\gamma'(t)) = \cos^2(t).$$

$$\int_0^1 \omega(\gamma'(t)) dt = \int_0^1 \cos^2(t) dt > 0$$

Now take the path $\Gamma : [0, \pi] \rightarrow \mathbb{R}^2$ defined by $\Gamma(t) = (\frac{t}{\pi}, 0)$

2 Tensors

Definition: Let $k, l \in \mathbb{N}$. The (k, l) -tensor bundle on M is defined by

$$T^{(k,l)}(M) = \underbrace{TM \otimes \cdots \otimes TM}_{k \text{ times}} \otimes \underbrace{T^*M \otimes \cdots \otimes T^*M}_{l \text{ times}}.$$

We call k the *contravariant degree* and l the *covariant degree*.

If $k = l = 0$, then

$$T^{(0,0)}(M) \cong M,$$

that is, smooth functions.

Contravariant components come from the tangent bundle and covariant components come from the cotangent bundle. The ordering convention is that contravariant indices appear first.

Examples:

1. $T^{(1,0)}(M) = TM$.
2. $T^{(0,1)}(M) = T^*M$.
- 3.

$$T^{(1,1)}(M) = TM \otimes T^*M \cong \text{Hom}(TM, TM).$$

So $(1, 1)$ -tensors can be viewed as linear endomorphisms of each tangent space.

Definition: A (k, l) -tensor field is a smooth section of $T^{(k,l)}(M)$.

2.1 Local coordinates description

Let (U, x) be a local chart with coordinates $(x^1, \dots, x^n)$ and let $p \in U$. Then

$$\left\{ \frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^n} \Big|_p \right\}$$

is a basis of $T_p M$, and

$$\left\{ dx^1 \Big|_p, \dots, dx^n \Big|_p \right\}$$

is the dual basis of $T_p^* M$.

A general element of $T_p^{(k,l)} M$ can be written as

$$\sum A_{j_1, \dots, j_l}^{i_1, \dots, i_k}(p) \frac{\partial}{\partial x^{i_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{i_k}} \otimes dx^{j_1} \otimes \dots \otimes dx^{j_l}.$$

The coefficients transform in a specific way under coordinate changes, this is what makes tensors coordinate-independent geometric objects.

2.2 Symmetric and antisymmetric tensors

Definition: Let $\alpha \in T^{(0,k)}(M)$. We say α is

1. Symmetric if

$$\alpha(v_1, \dots, v_k) = \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)})$$

for all permutations σ .

2. Antisymmetric if

$$\alpha(v_1, \dots, v_k) = (-1)^{\text{sign}(\sigma)} \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)}).$$

Antisymmetric tensors are the ones that will give rise to differential forms.

2.3 Pullback of covariant tensors

Let $F : M \rightarrow N$ be smooth and let $\alpha \in \Gamma(T^{(0,k)}(N))$.

Definition: The pullback $F^* \alpha \in \Gamma(T^{(0,k)}(M))$ is defined by

$$(F^* \alpha)(v_1, \dots, v_k) = \alpha(DF(v_1), \dots, DF(v_k)).$$

Notice that this only naturally works for covariant tensors. The differential pushes vectors forward, and the pullback compensates by feeding them into α .

3 Lie Derivative

Motivation: Let M and N be smooth manifolds of dimensions m and n , respectively. For any smooth map

$$F : M \longrightarrow N,$$

we can naturally associate a bundle map

$$dF : TM \longrightarrow TN,$$

often called the *global differential* of F . This construction generalizes the familiar differential of a smooth map

$$G : \mathbb{R}^m \longrightarrow \mathbb{R}^n,$$

and can be regarded as the derivative of F in the context of manifolds. The global differential is a standard concept that appears early in most introductory courses on differential geometry.

3.1 Why does the naive differential fail?

To understand the Lie derivative, we must first recognize why standard calculus cannot be directly applied to vector fields on manifolds.

The Global Differential Problem: Given a vector field Y on a manifold M , we can view it as a map $Y : M \rightarrow TM$. Its differential is $dY_p : T_p M \rightarrow T_{Y_p}(TM)$. This results in a vector in the tangent space of the tangent bundle, which is a $2n$ -dimensional space. We want a derivative that stays within the n -dimensional tangent space $T_p M$.

The Projection Obstruction: If we try to project this differential back down to M using the natural projection $\pi : TM \rightarrow M$, we find that $\pi \circ Y = \text{id}_M$. By the chain rule:

$$d\pi_{Y_p} \circ dY_p = d(\text{id}_M)_p = \text{id}_{T_p M}.$$

This means the “horizontal” part of the differential dY always yields the identity map, regardless of how the vector field Y actually behaves. It fails to capture the “change” in the field.

Disjointness of Tangent Spaces: The fundamental issue is that $T_p M$ and $T_q M$ are disjoint for $p \neq q$. In $\mathbb{R}^n$, we can subtract vectors at different points because there is a canonical way to identify all tangent spaces. On a manifold, there is no natural way to compare a vector at p with a vector at q without additional structure (like a flow or a connection).



Sophus Lie in 1896

3.2 Good way to do it! Geometric definition via flows

The Lie derivative $\mathcal{L}_X Y$ solves the comparison problem by using the **flow** θ_t of a vector field X to “transport” the vectors back to a single point.

Definition: Let θ_t be the flow of X . The Lie derivative of Y along X at point p is:

$$\mathcal{L}_X Y|_p = \lim_{t \rightarrow 0} \frac{(d\theta_{-t})_{\theta_t(p)}(Y_{\theta_t(p)}) - Y_p}{t}.$$

Basically we are “dragging” the vector field Y back from the point $\theta_t(p)$ to the point p using the differential of the flow, then perform the standard subtraction in $T_p M$.

3.3 The Lie bracket and coordinates

The Lie derivative is equivalent to the Lie bracket $[X, Y]$. In local coordinates x^i , if $X = X^i \partial_i$ and $Y = Y^j \partial_j$, the components are:

$$[X, Y]^j = X^i \frac{\partial Y^j}{\partial x^i} - Y^i \frac{\partial X^j}{\partial x^i}.$$

This can be interpreted as the commutator of the two vector fields acting as derivations on smooth functions:

$$[X, Y]f = X(Yf) - Y(Xf).$$

3.4 The Flow-Commutator Identity

A central result is the relationship between the Lie bracket and the commutator of the flows θ_t (for X) and ϕ_t (for Y).

Consider the curve

$$c(t) = (\phi_{-t} \circ \theta_{-t} \circ \phi_t \circ \theta_t)(p).$$

- The first derivative $c'(0)$ vanishes, which is why the second derivative acts as a derivation.
- The second derivative at $t = 0$ recovers the Lie bracket:

$$\left. \frac{d^2}{dt^2} \right|_{t=0} (\phi_{-t} \circ \theta_{-t} \circ \phi_t \circ \theta_t)(p) = 2[X, Y]_p.$$

Another way to think about this: The Lie derivative $\mathcal{L}_X Y$ can be formally understood as a second-order measure of the failure of the flows θ_t (associated with X) and ϕ_t (associated with Y) to commute. This relationship is captured by the *commutator curve*

$$c(t) = (\phi_{-t} \circ \theta_{-t} \circ \phi_t \circ \theta_t)(p).$$

Because the first-order Taylor terms of the respective flows cancel out, the first derivative $c'(0)$ vanishes. This uniquely allows the second derivative $c''(0)$ to act as a derivation (i.e., a tangent vector) on the manifold, rather than as a higher-order acceleration term. A second-order expansion reveals that the non-vanishing components of this “loop” are precisely the components of the Lie bracket, leading to the identity

$$c''(0) = 2[X, Y] = 2\mathcal{L}_X Y,$$

which interprets the Lie derivative as the infinitesimal displacement resulting from traversing a “square” defined by the two vector fields.

4 Differential forms

Definition: A differential k -form on M is a smooth section of

$$\wedge^k(T^*M).$$

We denote the space of all smooth k -forms by $\Omega^k(M)$.

We also define

$$\Omega^*(M) = \bigoplus_{k=0}^n \Omega^k(M).$$

A k -form is simply an antisymmetric covariant k -tensor field.

4.1 Local expression

In local coordinates $(x^1, \dots, x^n)$, a k -form can be written as

$$\alpha = \sum A_{i_1, \dots, i_k}(p) dx^{i_1} \wedge \dots \wedge dx^{i_k}.$$

The wedge product enforces antisymmetry. In particular, $dx^i \wedge dx^i = 0$.

If $\alpha \in \Omega^k(M)$ and $\beta \in \Omega^l(M)$, then

$$\alpha \wedge \beta \in \Omega^{k+l}(M).$$

4.2 Pullback of forms

If $F : M \rightarrow N$ is smooth and $\alpha \in \Omega^k(N)$, then

$$(F^*\alpha)(v_1, \dots, v_k) = \alpha(DF(v_1), \dots, DF(v_k)).$$

Pullback preserves wedge products and linearity.

In local coordinates, if

$$F(x^1, \dots, x^m) = (F^1(x), \dots, F^n(x)),$$

then

$$F^*(dy^i) = d(F^i) = \sum_{j=1}^m \frac{\partial F^i}{\partial x^j} dx^j.$$

Example:

1. Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by

$$F(x, y) = (F^1(x, y), F^2(x, y)).$$

Then

$$F^*(dx \wedge dy) = dF^1 \wedge dF^2.$$

Now compute

$$dF^1 = \frac{\partial F^1}{\partial x} dx + \frac{\partial F^1}{\partial y} dy, \quad dF^2 = \frac{\partial F^2}{\partial x} dx + \frac{\partial F^2}{\partial y} dy.$$

Therefore,

$$\begin{aligned} dF^1 \wedge dF^2 &= \left(\frac{\partial F^1}{\partial x} dx + \frac{\partial F^1}{\partial y} dy \right) \wedge \left(\frac{\partial F^2}{\partial x} dx + \frac{\partial F^2}{\partial y} dy \right) \\ &= \frac{\partial F^1}{\partial x} \frac{\partial F^2}{\partial x} dx \wedge dx + \frac{\partial F^1}{\partial x} \frac{\partial F^2}{\partial y} dx \wedge dy \\ &\quad + \frac{\partial F^1}{\partial y} \frac{\partial F^2}{\partial x} dy \wedge dx + \frac{\partial F^1}{\partial y} \frac{\partial F^2}{\partial y} dy \wedge dy. \end{aligned}$$

Since $dx \wedge dx = 0$ and $dy \wedge dy = 0$, and since

$$dy \wedge dx = -dx \wedge dy,$$

we obtain

$$dF^1 \wedge dF^2 = \left(\frac{\partial F^1}{\partial x} \frac{\partial F^2}{\partial y} - \frac{\partial F^1}{\partial y} \frac{\partial F^2}{\partial x} \right) dx \wedge dy.$$

Thus,

$$F^*(dx \wedge dy) = \det(JF) dx \wedge dy.$$

The antisymmetry of the wedge product is exactly what produces the determinant. This is not a coincidence, determinants measure oriented area scaling, and top-degree forms encode orientation. This explains why the Jacobian determinant appears in change of variables for integrals (it is a fundamental statement about how top-degree forms transform).

4.3 Exterior derivatives

Recall: Given $f \in C^\infty(M) = \Omega^0(M)$ $f : M \rightarrow \mathbb{R}$ you can take the differential $df \in \Omega^1(M)$.

The goal is to consider an arbitrary p -form and define its associated "differential" that gives you a $p+1$ -form.

Theorem: There exists a unique linear map

$$d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$$

satisfying the following:

1. For each $f \in C^\infty(M)$, $df(v) = v(f)$
2. $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$

Lemma: For a smooth map $F : M \rightarrow N$, $\alpha \in \Omega^k(N)$. Then

$$F^*(d\alpha) = d(F^*\alpha)$$

A general k -form looks like $fx + gdy + hdz$ in $\mathbb{R}^3$.

Lemma: For every $\alpha \in \Omega^k(M)$, $d(d\alpha) = 0$, i.e. $d^2 = 0$.

The previous lemma helps us to prove uniqueness of the operator d .

Example: $\alpha = f(x, y, z)dx + g(x, y, z)dy + h(x, y, z)dz$. Then

$$\begin{aligned} d\alpha &= df \wedge dx + dg \wedge dy + dh \wedge dz \\ &= (f_y dy + f_z dz) \wedge dx + (g_x dx + g_z dz) \wedge dy + (h_x dx + h_y dy) \wedge dz = \mathbf{curl} \end{aligned}$$

4.4 Interior derivative (inner derivation, contraction)

Theorem: Let V be a smooth vector field $V \in \Gamma(M)$ smooth vector field on M . Then there exists a unique linear map $\Omega^k(M) \rightarrow \Omega^{k-1}(M)$:

1. $V \lrcorner df = V(f)$
2. $V \lrcorner(\alpha \wedge \beta) = (V \lrcorner \alpha) \wedge \beta + (-1)^k \alpha \wedge (V \lrcorner \beta)$ for all $\alpha \in \Omega^k(M)$ and $\beta \in \Omega(M)$.

Note: $\alpha(V_1, V_2, \dots, V_k) = (V_k \lrcorner \dots (V_2 \lrcorner (V_1 \lrcorner \alpha)))$

Proposition: $\omega \in \Omega^1(M)$, $X, Y \in \Gamma(TM)$

$$d\omega(X, Y) = X(\omega(Y)) - Y(\omega(X)) - \omega([X, Y]).$$

5 Distributions and foliations

def smooth distribution.

Lemma: For all p in M , there exists a neighborhood U and smooth vector fields $X_1, \dots, X_k$ such that $X_1|_U, \dots, X_k|_U$ is a basis for $D|_U$ for all $U \subset M$. Conversely any subset $D \subset TM$ with this property is a rank k distribution.

Def: Given a rank k distribution $D \subset TM$ and integral submanifold of D is a submanifold $N \subset M$ st $T_p N = D|_p$ for all $p \in N$.

Examples

1. V nowhere vanishing vector field defines a rank one distribution $D_p = \text{span}(V(p))$ and the integral submanifold is integral curve or flow line.
2. $\{\partial/\partial x_1\}_1^k$ span a rank k distribution D in $\mathbb{R}^n$ with integral submanifold $\mathbb{R}^k$
3. $D = \text{Span}(X, Y)$, $X = \partial/\partial x + y\partial/\partial z$, $Y = \partial/\partial y$

Def: Involutive distribution

Def: A distribution $D \subset TM$ is integrable if each point $p \in M$ is contained in an integral submanifold of M

Prop: Every integral distribution is involutive.

Ex: Assume there is an integral submanifold $N \subset \mathbb{R}^3$

Lemma: A subset $D \subset TM$ is a rank k distribution iff for all $p \in M$ there exists $U \subset M$ and $\omega_1, \dots, \omega_{n-k} \in \Omega^1(M)$ s.t. for all $q \in U$

$$D|_q = \ker(\omega_1)|_q \cap \dots \cap \ker(\omega_{n-k})|_q$$

remark: the omegas are called defining 1-forms of the distribution.

def: A p -form ω annihilates D if for every open subset $U \subset M$ and for all $X_1, \dots, X_p \in \Gamma(D|_U)$

$$\omega(X_1, \dots, X_p) = 0$$

Lemma: Let $\omega_1, \dots, \omega_{n-k}$ be the defining 1 forms for a given rank k distribution D . A p form $\eta \in \Omega^p(M)$ annihilates D iff there exists $\beta_1, \dots, \beta_{n-k} \in \Omega^{p-1}(M)$ such that

$$\eta = \sum_{i=1}^{n-k} \omega_i \wedge \beta_i$$

Theorem: $D \subset TM$ distribution of codimension 1. Then D is involutive iff for every $U \subset M$ and any annihilating form $\eta \in \Omega^1(D|_U)$ we have that $d\eta$ also annihilates $D|_U$.

def: $M, D \subset TM$ rank k distribution. Then D is involutive if for all $U \subset M$ and for all $X, Y \in \Gamma(D|_U)$, $[X, Y] \in \Gamma(D|_U)$

Prop: $\Gamma(D) \subset \Gamma(TM)$. D is involutive iff $\Gamma(D)$ is a Lie subalgebra of $\Gamma(TM)$

Def: Let $M, D \subset TM$ is a rank k distribution. A local chart (U, φ) and local coordinates $(x_1, \dots, x_n)$ is flat for D if $\{\partial/\partial x_j\}$ is a frame for $D|_U$. D is completely integrable if $\forall p \in M$, p is contained in a chart (U, φ) which is flat for D .

Remark: D is completely integrable $\{x_{n-k} = c_{n-k}, \dots, x_n = c_n\}$ is a k dimensional integral submanifold of M .

Theorem Frobenius: Every involutive distribution is completely integrable.

Lemma: Let D be a rank k distribution. Suppose that $\forall p \in M$, there exists $U \subset M$ and a frame $X_1, \dots, X_k$ of $D|_U$ such that $[X_i, X_j] = 0$. Then D is completely integrable.